

Abstract

1 Introduction

Mobile phone surveys are an increasingly popular method for tracking household food security that is employed by aid organizations like the World Food Programme. Improving sampling capabilities and enabling real-time tracking of consumption and coping metrics, these surveys are conducted daily in an array of countries. Drawn with a random sampling strategy to select participants, samples do not necessarily overlap day to day, but are conducted at a frequency that often results in temporal correlation in the collected responses that hinders inference when using linear regression. While there already exist methods for analyzing data in which samples are surveyed repeatedly or aggregated at the time-step level, models that account for individual-level effects and population-level shocks are scarce. Those that do exist model temporal fluctuations with a latent time series that is recursively estimated and used to update parameter estimates via an EM algorithm, but these methods do not properly account for how cross-sectional noise impacts analysis. We correct this shortcoming by incorporating methods for analyzing data that are the result of multiple noisy observations of each element of a time series, known as an ARMA-Noise model.

We go on to establish that the use of an EM algorithm that factors the likelihood equation by conditioning on an estimate of a latent ARMA process incorrectly assumes that the resulting deviations will have a covariance structure conducive to standard fixed effect regression. We therefore replace the EM algorithm with one that recursively estimates variance parameters with an ARMA-Noise model and uses them to calculate GLS estimators for regression parameters. Confidence regions for time series parameters are generated via the Delta Method due to their latent nature. The respective implications of varying orders of the latent time ARMA process inform a decision rule for model selection.

While the newly incorporated method for fitting time series parameters corrects the bias that previously existed, it imposes a constant sample size at the time-step level, an unrealistic requirement in real-life surveying contexts. We therefore explored the extent to which ARMA-Noise modeling methods could be generalized to accommodate varying sample sizes.

With the proposed method fully developed, we test on simulated data and illustrate how model selection might be performed on a realization from a restricted model. We also include a case study on household survey data from Honduras that shows the method's potential to decorrelate residuals. The results of our analysis support the method as a viable option for modeling RCS survey data related to food security and other contexts.

2 Repeated Cross Sectional Data

Having detailed some useful machinery for analyzing univariate time series, we now consider data obtained through multiple observations, each with i.i.d noise, made at each time step. Such data are defined as repeated cross-sections (RCS), and they have long been the subject of analysis. When cross-sections are taken with high frequency, the possibility of temporal correlation arises. While panel data consist of a single sample being observed repeatedly, RCS data do not necessarily have any repeated observations, ruling out the treatment of the data as a multivariate time series. It is often instead the population mean that is treated as a time series, with individual observations modeled as a realization of that mean with some added noise.

3 Modeling RCS Data with a Univariate Time Series

Given such a dataset, it may seem apt to simply calculate averages at each time step and analyze these averages as a time series with covariates. Such a model is an example of a state-space model that can be modeled jointly via Kalman filtering [2]. In the regression setting, aggregates would be needed for both response and explanatory variables, fitting the following model:

$$\bar{\mathbf{y}} = (n^{-1} X_t' X_f) \beta + \mathbf{u}, \quad \mathbf{u} \sim \text{ARMA}(p, q)$$

This method is very effective at teasing out population-level fixed effects like intercepts and slopes, but is limited when it comes to the estimation of individual-level effects. A scenario in which this method falls short is the presence of indicator variables that vary little in their distribution at each time step. This is a highly plausible scenario when it comes to sampling methods for surveys. If researchers are interested in the effect of a certain binary covariate, for example, they may very well aim to be consistent in the proportions of time step samples that have this feature. Such consistency hinders the ability of the significant effect to be estimated as such, which we illustrate through a simulated simple regression model with a trend and autocorrelated errors:

$$y_{ti} = \beta_0 + \beta_1 t + \beta_2 x_{ti} + u_t + \varepsilon_{ti}, \quad i = 1, \dots, n_t$$

$$u_t - \phi u_{t-1} = \eta_t + \theta \eta_{t-1}, \quad t = 1, \dots, T$$

$$\eta_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\eta^2), \quad x_{ti} \sim \text{Bernoulli}(.5), \quad \varepsilon_{ti} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\varepsilon^2)$$

We consider fitting this model by aggregating the data at each time point and treating the averages as a time series with covariates. 100 simulations were conducted with the following parameters:

$$\beta = (43, -.001, .25), \quad \sigma_\varepsilon^2 = 100, \quad \phi = \theta = .5, \quad \sigma_\eta^2 = 1$$

Some sample rows of simulated data are provided below:

t	x	y
1.00	0	48.14
1.00	0	25.17
1.00	0	42.60
1.00	1	68.78
1.00	0	37.36
1.00	0	30.96

After aggregating at each time step, we obtain the following condensed dataset, from which a time series with covariates is then fit:

t	n	x	$\bar{y}$
1.00	94	0.37	44.88
2.00	84	0.50	43.24
3.00	96	0.47	43.49
4.00	107	0.52	42.68
5.00	89	0.47	42.45
6.00	99	0.49	41.87

Note that the x column of each time step sample has a value close to 0.5, with some variability. If we then fit a time series with covariates to the condensed dataset, we obtain results with empirical means displayed in the table below. The “Significance Frequency” column displays the fraction of the hundred simulations in which the corresponding parameter estimate was statistically significant at the $\alpha = .05$ level.

Parameter	True Value	Estimate Mean	Sig. Freq.
phi	0.500	0.494	1.000
theta	0.500	0.001	0.040
Intercept	43.000	43.017	1.000
Slope	-0.001	-0.001	0.860
Covariate	0.250	0.216	0.110

Note that the effect of the dummy indicator variable is deemed significant only 11% of the time. This example highlights the shortcomings of fitting a time series with covariates. Also of note is how skewed the sample mean of $\hat{\theta}$ is, in particular how close to zero it is. We will see later why this is actually expected. For now, we use these results to motivate the need for a different approach. With that need in mind, we next review some of the key developments in modeling repeated cross-sectional data.

4 Early Methods for Analyzing RCS Data

Among the earliest methods designed for RCS data was that proposed by Scott and Smith [6], in which dependent variables are aggregated at the time step level and modeled as a time series with added noise:

$$y_t = \theta_t + e_t, \quad \text{Var}(e_t) = s_t^2 \quad (4.1)$$

In this model, surveys are conducted at regular time intervals and θ_t represents a population mean changing over time. The analytical method proposed therein introduces different possibilities for the covariance structures of $\{\theta_t\}$ and $\{e_t\}$, but is ultimately concerned with the use of all previously observed aggregates $\{y_t\}_{t=1}^T$ to estimate θ_T . As our objective is to improve the estimation and of covariate effects and their standard errors, we must look to more highly parameterized models.

Such a model was developed by DiPrete and Grusky [3], in which added complexity and flexibility is added to a standard fixed effects model by having the effects of individual-level variables depend on macro-level variables:

$$y_{ti} = X_{ti}\beta_t + \varepsilon_{ti}, \quad \beta_t = \mathbf{Z}_t\gamma, \quad \varepsilon_{ti} \sim \text{WN}(0, \sigma^2) \quad (4.2)$$

Such a model requires the presence of macro-level variables that vary over time and explain changing effects of individual-level variables. While this formulation has a broad range of applications, it is only appropriate in cases where temporal correlation can be assumed to be a function of known time-varying predictors, rather than an independent random process.

5 Autoregressive Random Effects Models

One model that has been developed to analyze RCS data with autocorrelated errors is the Autoregressive Random Effects (ARE) Model [5]. As it is the model that served as our starting point, we describe it in greater detail. Treating observations as the sum of a fixed effects model and an AR(1) time series, ARE models have two levels:

$$y_{ti} = \mathbf{x}_{ti}\beta + u_t + \epsilon_{ti}, \quad i = 1, \dots, n, \quad \epsilon_{ti} \sim \text{i.i.d. } N(0, \sigma_\epsilon^2) \quad (5.1)$$

$$u_t - \phi u_{t-1} = \eta_t + \theta \eta_{t-1}, \quad t = 1, \dots, T, \quad \eta_t \sim N(0, \sigma_\eta^2) \quad (5.2)$$

A realization of an ARE model is displayed in Figure 1 with the following parameters:

Parameter	β_0	β_t	β_1	σ_ϵ^2	ϕ	σ_η^2
Value	40	-0.1	2	1	0.9	1

Table 1: Parameter values for simulated ARE model

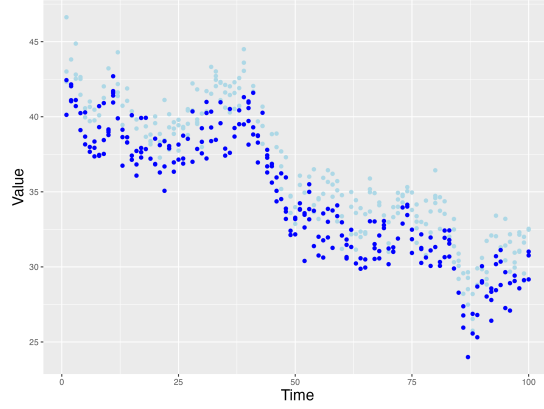


Figure 1: Realization of ARE model with parameters defined in Table 1

Setting $N := nT$, the total number of observations in the dataset, we define the fixed effect and temporal fluctuation design matrices X_f and X_t (resp.) to be:

$$X_f = \begin{bmatrix} \mathbf{1}_n & \mathbf{1}_n & \mathbf{x}_1 \\ \mathbf{1}_n & \mathbf{2}_n & \mathbf{x}_2 \\ \vdots & \vdots & \vdots \\ \mathbf{1}_n & \mathbf{T}_n & \mathbf{x}_T \end{bmatrix}, \quad X_t = \begin{bmatrix} \ddots & & \mathbf{0} \\ & \mathbf{1}_n & \\ \mathbf{0} & & \ddots \end{bmatrix}$$

With these matrices we can concisely represent the model governing the vector of observations $\mathbf{y}$:

$$\mathbf{y} = X_f \beta + X_t \mathbf{u} + \varepsilon$$

The method employed to fit such a model is a recursive process that splits the log likelihood function of the model into two terms and then utilizes an EM algorithm for parameter estimates. We introduce some necessary notation for expressing the likelihood function of the full set of parameters:

$$\mathbb{R}^{T \times T} \ni \Gamma := \text{Var}(\mathbf{u}), \quad \Gamma_{i,j} = \gamma_{\mathbf{u}}(i-j) = \frac{\phi^{|i-j|} \sigma_{\eta}^2}{1 - \phi^2} \quad (5.3)$$

Note that $\gamma_{\mathbf{u}}(h)$ always has a factor of σ_{η}^2 , meaning the Toeplitz matrix $\mathbf{V}$ can be defined such that $\Gamma = \sigma_{\eta}^2 \mathbf{V}$:

$$\mathbf{V} = \frac{1}{1 - \phi^2} \begin{bmatrix} 1 & \phi & \dots & \phi^{T-1} \\ \phi & 1 & \ddots & \phi^{T-2} \\ \vdots & \ddots & \ddots & \vdots \\ \phi^{T-1} & \phi^{T-2} & \dots & 1 \end{bmatrix} \quad (5.4)$$

We can also use X_t to concisely represent the covariance matrix of $\mathbf{y}$:

$$\mathbb{R}^{N \times N} \ni \Omega := \text{Var}(\mathbf{y}) = X_t \mathbf{\Gamma} X_t' + \sigma_\varepsilon^2 I_N \quad (5.5)$$

We define ξ to be the vector of all parameters $(\beta, \sigma_\varepsilon^2, \phi, \sigma_\eta^2)$. The vector $\mathbf{y}$ is a multivariate Gaussian random variable with mean $X_f \beta$, variance Ω , and the following log likelihood function:

$$l(\xi|\mathbf{y}) = \frac{-N}{2} \ln(2\pi) - \frac{1}{2} \ln|\Omega| - \frac{1}{2} (\mathbf{y} - X_f \beta)' \Omega^{-1} (\mathbf{y} - X_f \beta) \quad (5.6)$$

By conditioning on $\mathbf{u}$, we can split the log likelihood function into two levels:

$$l(\xi|\mathbf{y}) = l_1(\xi_1|\mathbf{y}, \mathbf{u}) + l_2(\xi_2|\mathbf{u}), \quad \xi_1 = (\beta, \sigma_\varepsilon^2), \quad \xi_2 = (\phi, \sigma_\eta^2)$$

$$l_1(\xi_1|\mathbf{y}, \mathbf{u}) = \ln(f_{Y|U}(\mathbf{y}|\mathbf{u}, \xi_1)) = -\frac{N}{2} \ln(2\pi\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2} \|\mathbf{y} - X_t \mathbf{u} - X_f \beta\|^2 \quad (5.7)$$

$$l_2(\xi_2|\mathbf{u}) = \ln(f_U(\mathbf{u})) = -\frac{T}{2} \ln(2\pi\sigma_\eta^2) + \frac{1}{2} \ln(1 - \phi^2) - \frac{1}{2\sigma_\eta^2} \mathbf{u}' V^{-1} \mathbf{u} \quad (5.8)$$

This conditional splitting is convenient, as $\mathbf{u} \sim N(\mathbf{0}, \mathbf{\Gamma})$ and $\mathbf{y}|\mathbf{u} \sim N(X_f \beta + \mathbf{u}, \sigma_\varepsilon^2 I_N)$. Furthermore, the choice of imposing an AR(1) structure on $\mathbf{u}$ lends itself to a concise representation of the determinant and inverse of $\mathbf{\Gamma}$:

$$\mathbf{V}^{-1} = \begin{bmatrix} 1 & -\phi & 0 & \cdots & 0 & 0 \\ -\phi & 1 + \phi^2 & -\phi & \cdots & 0 & 0 \\ 0 & -\phi & 1 + \phi^2 & \ddots & 0 & 0 \\ \vdots & & & \ddots & \ddots & -\phi & 0 \\ 0 & \cdots & 0 & -\phi & 1 + \phi^2 & -\phi \\ 0 & 0 & \cdots & & -\phi & 1 \end{bmatrix} \quad (5.9)$$

$$\mathbf{\Gamma}^{-1} = \sigma_\eta^{-2} \mathbf{V}^{-1}, \quad |\mathbf{\Gamma}| = \sigma_\eta^{2T} (1 - \phi^2) \quad (5.10)$$

In the absence of an explicit observation of $\mathbf{u}$, its estimate is repeatedly calculated and used to update parameter estimates. Its estimation is therefore a key part of this algorithm, and is given by its expectation, conditioned on $\mathbf{y}$:

$$\mathbb{E}(\mathbf{u}|\mathbf{y}) = \mathbf{\Gamma} X_t' \Omega^{-1} (\mathbf{y} - X_f \beta) \quad (5.11)$$

This estimate $\hat{\mathbf{u}}$ will have the following covariance structure:

$$\text{Var}(\hat{\mathbf{u}}) = \Sigma_{\hat{\mathbf{u}}, \hat{\mathbf{u}}} = \sigma_\eta^2 \mathbf{\Gamma} X_t' \Omega^{-1} X_t \mathbf{\Gamma} \quad (5.12)$$

With this estimate now defined, we summarize the fitting of an ARE model:

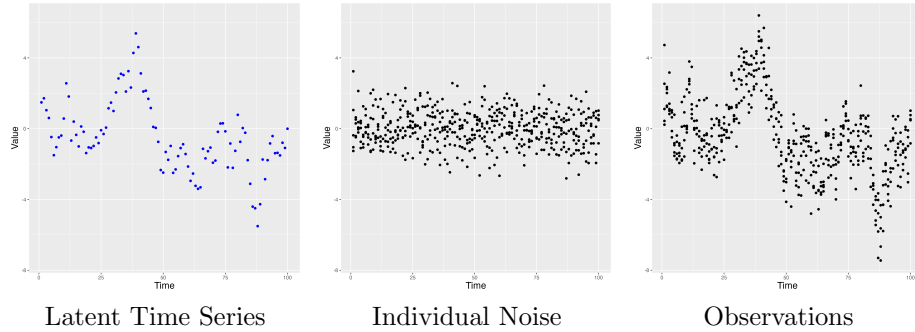
- 1) Calculate initial parameter estimates;
- 2) Estimate $\mathbf{u}$ with $\mathbb{E}(\mathbf{u}|\mathbf{y})$;
- 3) Apply $\hat{\mathbf{u}}$ with an AR(1) likelihood to update $\hat{\xi}_2$ and fit $\mathbf{y} - X_t\hat{\mathbf{u}}$ with a fixed effects model to update $\hat{\xi}_1$;
- 4) Calculate and store $l_1(\hat{\xi}_1)$ and $l_2(\hat{\xi}_2)$;
- 5) Repeat steps 2-4 until convergence criteria are met.

All estimate updates are based on explicitly calculable maximum likelihood estimators, with the exception of the AR parameter ϕ , which is updated via one iteration of the Newton-Raphson scheme since l_2 is not linear with respect to ϕ . In the next section, we examine some potential issues with fitting models with the ARE approach.

6 How Additional White Noise Alters ARMA Processes

Let's assume the best case scenario for level one of the model, namely the perfect estimation of fixed effect parameters ($\hat{\beta} = \beta$). In this unlikely event, we are able to take data such as that displayed in Figure 1 and extract $\mathbf{w} := X_t\mathbf{u} + \varepsilon$, which is a realization of an *ARMA-Noise Model* [4]. This decomposition is displayed in Figure 2:

Figure 2: A realization of a repeatedly observed AR(1) time series with added noise



With no prior information about the latent time series, the best estimates for the elements of $\mathbf{u}$ are the averages at each time step:

$$\hat{\mathbf{u}} = n^{-1} X_t'(X_t\mathbf{u} + \varepsilon) = \mathbf{u} + \mathbf{a} =: \bar{\mathbf{w}}, \quad a_t = n^{-1} \sum_{i=1}^n \varepsilon_{ti}$$

As is shown in Figure 3, the daily averages (in red) seem to follow the true latent time series (in blue) fairly closely. We will soon see, however, that the added noise vector $\mathbf{a}$ introduces bias in the estimation of the time series parameters. To describe this bias, let's compare the respective ACVF's of $\mathbf{u}$ and $\hat{\mathbf{u}}$.

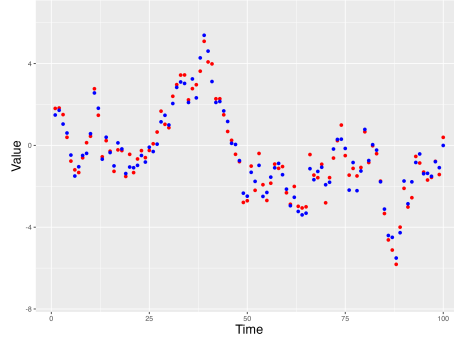


Figure 3: A Sample Latent Time Series (blue) and Daily Averages of Observed Data (red)

Being an AR(1) process, the covariance of u_t and $u_{t \pm h}$ is given by Equation ???. By virtue of the independence of $\{\eta_t\}$ and $\{a_t\}$, the ACVF of $\hat{\mathbf{u}}$ is affected as follows:

$$\gamma_{\hat{\mathbf{u}}}(h) = \begin{cases} \gamma_{\mathbf{u}}(0) + \frac{\sigma_{\varepsilon}^2}{n}, & h = 0 \\ \gamma_{\mathbf{u}}(h), & |h| > 0 \end{cases} \quad (6.1)$$

The fact that $\frac{\gamma_{\mathbf{u}+\mathbf{a}}(1)}{\gamma_{\mathbf{u}+\mathbf{a}}(0)} \neq \frac{\gamma_{\mathbf{u}+\mathbf{a}}(2)}{\gamma_{\mathbf{u}+\mathbf{a}}(1)}$ precludes $\hat{\mathbf{u}} = \mathbf{u} + \mathbf{a}$ from being an AR(1) process, but as we will now show, its ACVF is compatible with that of an ARMA(1,1). From Box (1976) [1], we have the following theorem:

If $\{y_t\}$ follows an ARMA(p_1, q_1) model, $\{a_t\}$ follows an ARMA(p_2, q_2) and the two series are independent, then their sum $w_t = y_t + a_t$ follows an ARMA(p, q) model in which $p \leq p_1 + p_2$ and $q \leq \max(p_1 + q_2, p_2 + q_1)$.

Because $\{a_t\}$ is white noise, $p_2 = q_2 = 0$. As $p_1 = 1$ and $q_1 = 0$, we can deduce that $\{w_t\}$ is an ARMA(p, q) process where $p \leq 1$ and $q \leq 1$. Because $\gamma_{\hat{\mathbf{u}}}(h+1)/\gamma_{\hat{\mathbf{u}}}(h) = \phi$ for $h > 0$, we know that $p = 1$. The ACVF of $\hat{\mathbf{u}}$ is therefore incompatible with that of an AR(1) process, so it must be ARMA(1,1).

The fact that both ACVF's are altered by a factor of ϕ after lag 1 means that the AR component is unaffected by the addition of white noise. While noting the AR term ϕ goes unchanged by the added noise, the estimation thereof will be irredeemably biased if fit with an AR(1) model (See Figure 4). $\xi_2 = (\phi, \sigma_{\eta}^2)$ must instead be estimated with a method that accounts for this additional noise

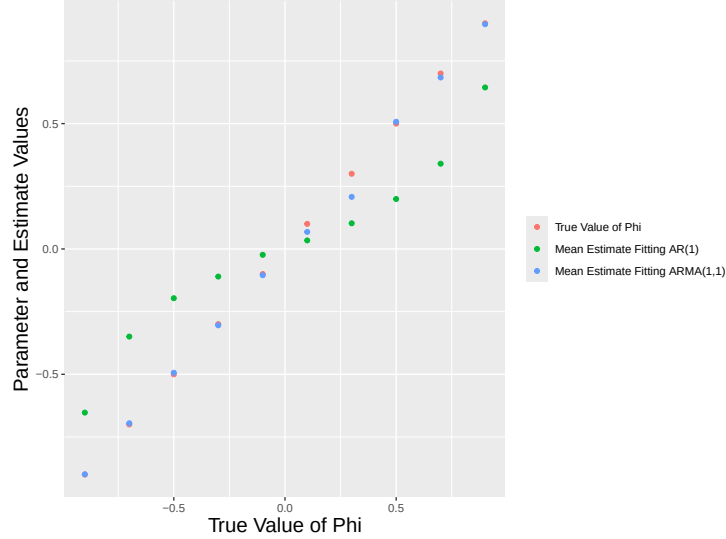


Figure 4: Average estimates for ϕ after fitting $\hat{\mathbf{u}} = \mathbf{u} + \mathbf{a}$ with an AR(1) model and an ARMA(1,1) model. The results were produced by 100 realizations with $T = 1000$, $\sigma_\eta^2 = \sigma_a^2$. The disparity between the accuracy of the two means grows with the relative magnitude of σ_a^2 with respect to σ_η^2 .

7 Analyzing ARMA-Noise Processes

Remaining in our best-case scenario of perfectly estimating β , our objective is to fit estimates for $\phi, \theta, \sigma_\eta^2, \sigma_\varepsilon^2$ with observations $\{w_{ti}\}$, the time-step averages of which form a time series $\bar{\mathbf{w}}$ that is parameterized by $\phi, \theta_b, \sigma_b^2$ (with innovations $\{b_t\}$). ARMA-Noise models are fit with a method detailed by Wong and Miller (1990) [4].

8 Impacts of time series estimation error on confidence regions for regression parameters

We’ve already looked at the implications of perfectly estimating β to improve how Level 2 parameters are estimated. Let’s now go in the other direction, assuming perfect estimation of ξ_2 to investigate the implications on fitting $\hat{\beta}$. This hypothetical is somewhat contrived, as we could only isolate $X_t\mathbf{u} + \varepsilon$ in the event that β was estimated perfectly, in which case we would have no need to estimate it again. To that end, let’s assume that our algorithm of choice perfectly estimates β on a particular update, but doesn’t “know” it. It then perfectly fits $\hat{\xi}_2$ using the method outlined by Wong, and the question becomes whether β is a “stable” estimate.

In the event that ξ_2 is perfectly estimated, all parameter estimates $\hat{\xi}$ are

plugged into the conditional expectation formula given in Equation 5.11. Even with perfect parameter estimates, $\hat{\mathbf{u}}$ will still deviate from $\mathbf{u}$, as $\text{Var}(\hat{\mathbf{u}} - \mathbf{u}) = \mathbf{\Gamma} - \mathbf{\Gamma}X_t'\Omega^{-1}X_t\mathbf{\Gamma}$. As a result, $\widehat{X_f\beta + \varepsilon} = \mathbf{y} - X_t\hat{\mathbf{u}}$ will not be the sum of fixed effects and white noise as the current algorithm supposes. Were we to treat $\mathbf{y} - X_t\hat{\mathbf{u}}$ as such, our resulting estimate for β would be the standard OLS estimate $\hat{\beta} = (X_f'X_f)^{-1}X_f'(\mathbf{y} - X_t\hat{\mathbf{u}})$

However, the inevitable error in $\hat{\mathbf{u}}$ would endow $\hat{\beta}$ not with the implied $\text{Var}(\hat{\beta}) = (X_f'X_f)^{-1}\sigma_\varepsilon^2$, but rather the following covariance matrix:

$$\text{Var}(\hat{\beta}) = 2\sigma_\varepsilon^2(X_f'X_f)^{-1} - \sigma_\varepsilon^2(X_f'X_f)^{-1}X_f'(\Omega^{-1})X_f(X_f'X_f)^{-1} =: \Sigma_\beta \quad (8.1)$$

As convenient as it might be to subtract corresponding entries of $\hat{\mathbf{u}}$ from $\mathbf{y}$ and treat the resulting difference as a fixed effects model with independent noise, the resulting confidence regions would be incorrect.

As the best-case scenario of estimating and extracting the latent time series will still yield temporal correlation in the data, albeit reduced, we must turn to a more nuanced treatment of residual covariance. A natural choice is the GLS estimator of β , which we will see challenges the necessity of using $\hat{\mathbf{u}}$ in the estimation of β at all.

As we now have a viable method (courtesy of Wong and Miller) for estimating all the parameters necessary to estimate Ω , an analog of the above equation with sample estimates will be our estimator for β , and confidence intervals will be based on a sample estimate of its covariance matrix $\text{Var}(\hat{\beta}_{\text{GLS}}) = (X_f'\Omega^{-1}X_f)^{-1}$. With all these considerations in mind, we are now ready to propose our method for fitting these types of models.

9 Methods

9.1 Incorporating ARMA-Noise into EM Algorithm

With the analysis of ARMA-Noise processes already established [4], we examine how such an analysis may be applied, not to explicit realizations of ARMA-Noise processes, but rather estimates of one in the form of residuals from a fixed effects model. We define our model similarly to the ARE model, but with the latent time series $\mathbf{u}$ now specified as an ARMA(1,1) model as opposed to strictly AR(1):

$$y_{ti} = \mathbf{x}_{ti}\beta + u_t + \epsilon_{ti}, \quad i = 1, \dots, n \quad \epsilon_{ti} \sim \text{i.i.d. } N(0, \sigma_\epsilon^2) \quad (9.1)$$

$$u_t - \phi u_{t-1} = \eta_t + \theta \eta_{t-1}, \quad t = 1, \dots, T \quad \eta_t \sim N(0, \sigma_\eta^2) \quad (9.2)$$

The algorithm we propose is an iterative process applying the estimates of parameters from one level to update those of the other. The process begins similarly to that outlined by Modugno *et al*, with a few key differences:

- 1) Estimate β with $\hat{\beta} = (X_f' X_f)^{-1} X_f' \mathbf{y}$;
- 2) Estimate $X_t \mathbf{u} + \varepsilon$ with $\mathbf{y} - X_f \hat{\beta}$ and fit ARMA-Noise model to obtain estimates $(\hat{\phi}, \hat{\theta}, \hat{\sigma}_\eta^2, \hat{\sigma}_\varepsilon^2)$;
- 3) Calculate $\hat{\Omega}$ and $l(\hat{\xi})$;
- 4) Refit fixed effect parameter estimates by calculating GLS estimate $\hat{\beta}_{GLS} = (X_f' \hat{\Omega}^{-1} X_f)^{-1} X_f' \hat{\Omega}^{-1} \mathbf{y}$;
- 5) Repeat Steps 2-4 until convergence criteria are met.;
- 6) Generate confidence regions and eliminate insignificant parameters if applicable.

9.2 Evaluating Convergence

Instead of splitting the likelihood by conditioning on $\hat{\mathbf{u}}$, doing so with $\bar{\mathbf{y}}$ instead of has the advantage that likelihoods are based on a constant statistic, as opposed to the ever-changing estimate of $\mathbf{u}$.

Let's recall the log likelihood $l(\xi)$:

$$l(\xi) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln|\Omega| - \frac{1}{2} (\mathbf{y} - X_f \beta)' \Omega^{-1} (\mathbf{y} - X_f \beta)$$

We now condition on $\bar{\mathbf{y}}_t = n^{-1} X_t' \mathbf{y}$, which has the following distribution

$$\bar{\mathbf{y}} \sim N(n^{-1} X_t' X_f \beta, n^{-2} X_t' \Omega X_t)$$

After simplifying, we get the following expression for $\Sigma_{\bar{\mathbf{y}}}$:

$$\Sigma_{\bar{\mathbf{y}}} = \Gamma + n^{-1} \sigma_\varepsilon^2 I_T,$$

which has the following log likelihood:

$$l_t(\xi) = \ln[f(\bar{\mathbf{y}}|\xi)] = -\frac{T}{2} \ln(2\pi) - \frac{1}{2} \ln|\Sigma_{\bar{\mathbf{y}}}| - \frac{1}{2} (\bar{\mathbf{y}} - n^{-1} X_t' X_f \beta)' \Sigma_{\bar{\mathbf{y}}}^{-1} (\bar{\mathbf{y}} - n^{-1} X_t' X_f \beta)$$

If we then condition $\mathbf{y}$ on $\bar{\mathbf{y}}$, the only randomness left is in the vector of deviations, which are independent of $\bar{\mathbf{y}}$:

$$\mathbf{y} - X_t \bar{\mathbf{y}} = \boldsymbol{\varepsilon} - X_t \mathbf{a}$$

Let's establish the distribution of this vector of deviations. The mean is immediately zero, and the covariance is given by:

$$\text{Var}(\boldsymbol{\varepsilon} - X_t \mathbf{a}) = \text{Var}(\boldsymbol{\varepsilon}) - 2\text{Cov}(\boldsymbol{\varepsilon}, X_t \mathbf{a}) + \text{Var}(X_t \mathbf{a})$$

which reduces to :

$$\text{Var}(\varepsilon - X_t \mathbf{a}) = \sigma_\varepsilon^2 I_N - 2\sigma_\varepsilon^2 n^{-1} X_t X_t' + n^{-2} \sigma_\varepsilon^2 (X_t X_t')^2$$

Because $X_t' X_t = n I_N$, we get:

$$\text{Var}(\varepsilon - X_t \mathbf{a}) = \sigma_\varepsilon^2 (I_N - n^{-1} X_t X_t') \quad (9.3)$$

We now derive the second part of our split likelihood. Because of the rank-deficient nature of the matrix Equation 9.3, we can simplify the log likelihood

$$l_i(\xi) = \ln f(\mathbf{y} \mid \bar{\mathbf{y}}, \sigma_\varepsilon^2) = -\frac{N-T}{2} \ln(2\pi\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2} \sum_{t=1}^T \sum_{i=1}^n (y_{it} - \bar{y}_t)^2, \quad (9.4)$$

With this we can express the full likelihood as $l(\xi) = l_t(\xi) + l_i(\sigma_\varepsilon^2)$. However, since $\bar{\mathbf{y}}_t$ is independent of our model selection and parameter estimates, l_i is going to be relatively stable. We therefore pair $\xi \in \mathbb{R}^k$ and $l_t(\xi)$ to calculate the Akaike Information Criterion (AIC):

$$AIC = -2k - 2l_t(\xi)$$

Calculating this weighted likelihood is how we will compare the performance of models that include different subsets of parameters.

10 Inference

$$\sqrt{T} \left[\begin{pmatrix} \hat{\phi} \\ \hat{\theta}_b \\ \hat{\sigma}_b^2 \end{pmatrix} - \begin{pmatrix} \phi \\ \theta_b \\ \sigma_b^2 \end{pmatrix} \right] \xrightarrow{d} N(\mathbf{0}, \Sigma_b)$$

$$\Sigma_b = \begin{bmatrix} \frac{(1-\phi^2)(1+\phi\theta_b)^2}{(\phi+\theta_b)^2} & \frac{-(1-\theta_b^2)(1-\phi^2)(1+\phi\theta_b)}{(\phi+\theta_b)^2} & 0 \\ \frac{-(1-\theta_b^2)(1-\phi^2)(1+\phi\theta_b)}{(\phi+\theta_b)^2} & \frac{(1-\theta_b^2)(1+\phi\theta_b)^2}{(\phi+\theta_b)^2} & 0 \\ 0 & 0 & 2\sigma_b^4 \end{bmatrix}$$

We also have the asymptotic distribution of the normalized estimate for σ_ε^2 :

$$\sqrt{T}(\hat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2) \xrightarrow{d} N(0, \frac{2\sigma_\varepsilon^4}{n-1})$$

Because $y_{t.} \perp S_t^2$ for all t , we know that $\hat{\sigma}_\varepsilon^2 \perp (\hat{\phi}, \hat{\theta}_b, \hat{\sigma}_\eta^2)$ and can give the following asymptotic distribution:

$$\sqrt{T}(\hat{\tau}_o - \tau_o) \xrightarrow{d} N(\mathbf{0}, \Sigma_o), \quad \Sigma_o = \begin{bmatrix} \Sigma_b & \mathbf{0} \\ \mathbf{0} & \frac{2\sigma_\varepsilon^2}{n-1} \end{bmatrix} \quad (10.1)$$

Equipped with the above estimates, we would then solve H for estimates of θ and σ_η^2 . It is important to note that estimates for τ_o can produce a system of equations that has no real roots, especially for relatively low values of T . In

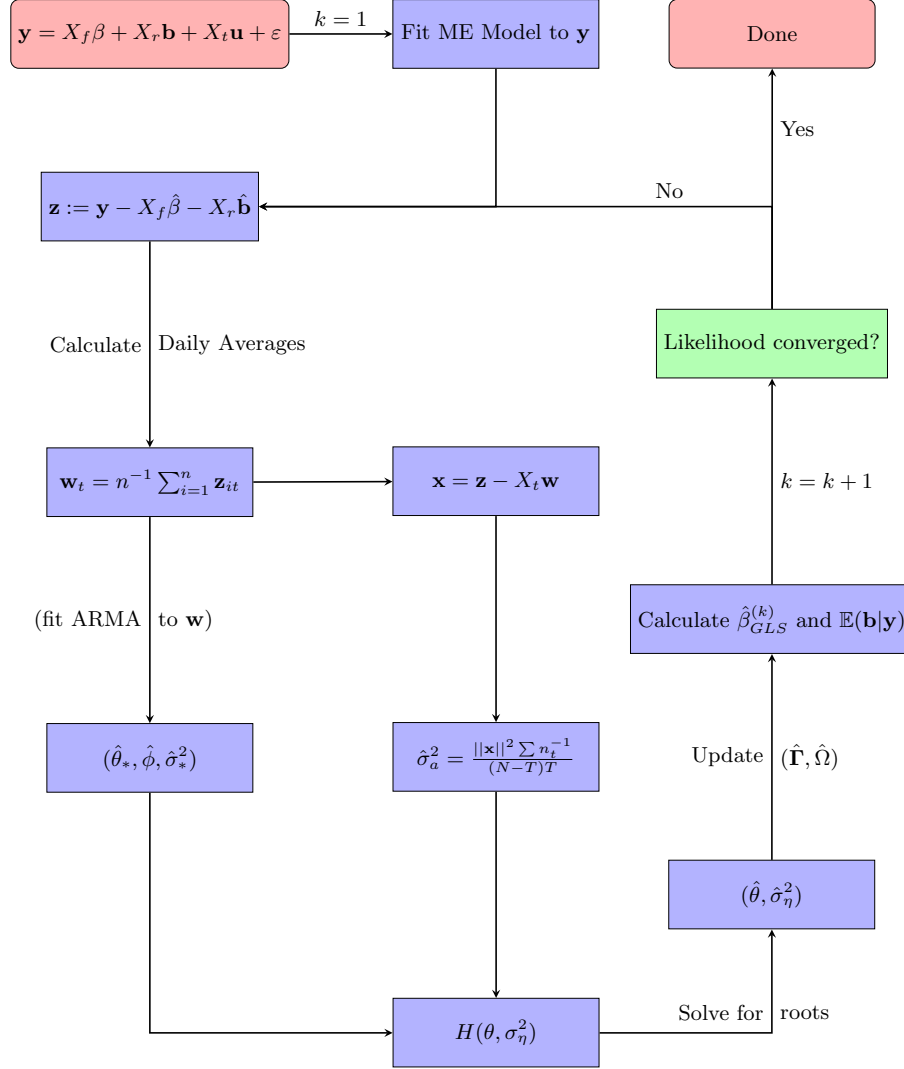


Figure 5: Flowchart of algorithm steps for estimating model parameters

this event we use numerical methods to solve for parameter estimates $\hat{\theta}, \hat{\sigma}_\eta^2$ that minimize $\|H\|^2$.

Were we to have an explicit function $(\hat{\phi}, \hat{\theta}, \hat{\sigma}_\eta^2) = F(\hat{\tau}_o)$ providing estimates for τ_l , we would apply the Delta Method to obtain the asymptotic distribution of our vector of parameter estimates:

$$\sqrt{T}(F(\hat{\tau}_o) - \tau_l) \xrightarrow{d} N(\mathbf{0}, G\Sigma_o G'), \quad G = \frac{\partial F}{\partial(\tau_o)} \quad (10.2)$$

Because we're using H , a function whose roots are our estimates, we will adapt 10.2 using the Implicit Function Theorem:

$$G = - \left(\frac{\partial H_f}{\partial \tau_1} \right)^{-1} \left(\frac{\partial H}{\partial \tau_0} \right) \quad (10.3)$$

Because we're ultimately interested in the normalized asymptotic distribution of parameter estimates $(\hat{\phi}, \hat{\theta}, \hat{\sigma}_\eta^2, \hat{\sigma}_\varepsilon^2)$, we expand H to include some identity functions.

Our procedure is to fit $\hat{\tau}_o$, solve for $\hat{\tau}_l$, and then plug in all estimates into Equations 10.2 and 10.3

$$\tau_0 = \begin{pmatrix} \phi \\ \theta_b \\ \sigma_b^2 \\ \sigma_\varepsilon^2 \end{pmatrix} \quad \tau_1 = \begin{pmatrix} \phi_* \\ \sigma_{\varepsilon*}^2 \\ \theta \\ \sigma_\eta^2 \end{pmatrix} \quad (10.4)$$

In verifying that the estimation of these parameters proceeds as expected, we can utilize the distribution of the following statistic:

$$T(\hat{\tau}_o - \tau_o)' \Sigma_o^{-1} (\hat{\tau}_o - \tau_o) \stackrel{d}{=} \chi_3^2 \quad (10.5)$$

We can therefore test whether estimates of the parameters of an ARMA-Noise process fall within predicted confidence regions for known parameters.

11 Model Selection

Before outlining the methodology for model selection, we motivate the procedure by exploring at the implications of setting certain parameters in an ARMA-Noise model $\bar{\mathbf{w}}_t = \mathbf{u} + \mathbf{a}$ equal to zero.

11.0.1 Case 1: $\mathbf{u}$ is AR(1)

We have already shown that this case results in $\mathbf{w}$ being an ARMA(1,1) process with innovations $\{b_t\}$ and parameterized by $(\phi, \theta_b, \sigma_b^2)$. Figures ?? and ?? remind us that θ_b will fall somewhere between 0 and $-\phi$, converging to the former as $\frac{\sigma_a^2}{\sigma_\eta^2}$ approaches 0 and the latter as it approaches infinity.

Assuming values of ϕ , σ_a^2 , and T of large enough magnitude to result in significant estimates for ϕ and θ_b , and not knowing $\mathbf{u}$ is AR(1), we use Equations

??, 10.2, and 10.3 to estimate and perform inference on $\hat{\tau}_l$. In the event $\hat{\theta}$ is insignificant we impose $\theta = 0$ and re-estimate σ_η^2 using H_θ .

$$H_\theta = (1 + \theta_b^2)\sigma_b^2 - \sigma_\eta^2 - (1 + \phi^2)\sigma_a^2, \quad H_f = \begin{cases} \sigma_{\varepsilon*}^2 - \sigma_\varepsilon^2 \\ \phi_* - \phi \\ H_\theta \end{cases} \quad (11.1)$$

$$\Sigma_o = \begin{bmatrix} \Sigma_{\phi, \theta_b} & \mathbf{0} & \mathbf{0} \\ \mathbf{0}' & 2\sigma_b^4 & 0 \\ \mathbf{0}' & 0 & \frac{2\sigma_\varepsilon^4}{n-1} \end{bmatrix} \quad (11.2)$$

$$\tau_o = (\phi, \theta_b, \sigma_b^2, \sigma_\varepsilon^2)', \quad \tau_l = (\sigma_{\varepsilon*}^2, \phi_*, \sigma_\eta^2)'$$

$$\frac{\partial H_f}{\partial \tau_o} = \begin{bmatrix} 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ -2\phi\sigma_a^2 & 2\theta_b\sigma_b^2 & (1 + \theta_b^2) & (1 + \phi^2)n^{-1} \end{bmatrix}, \quad \frac{\partial H_f}{\partial \tau_l} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

11.0.2 Case 2: $\mathbf{u}$ is MA(1)

We now consider the case where $\phi = 0$. If $\mathbf{u}$ is MA(1), so too is $\bar{\mathbf{w}}_t$, with parameters θ_b and σ_b^2 . The resulting system of related ACVF's is:

$$(1 + \theta_b B)b_t = (1 + \theta B)\eta_t + a_t$$

With this restricted relation we adapt our implicit function:

$$H_\phi = \begin{cases} (1 + \theta_b^2)\sigma_b^2 - (1 + \theta^2)\sigma_\eta^2 - \sigma_a^2 \\ \theta_b\sigma_b^2 - \theta\sigma_\eta^2 \end{cases}$$

Because the added noise in no way alters the lack of auto-regression, fitting an ARMA(1,1) process to $\mathbf{w}$ should yield an insignificant estimate for ϕ . In this event, $\mathbf{w}$ would be refit with an MA(1) model before estimating θ and σ_η^2 with an adjusted version of H , which we will denote H_ϕ :

$$H_\phi = \begin{cases} (1 + \theta_b^2)\sigma_b^2 - (1 + \theta^2)\sigma_\eta^2 - \sigma_a^2 \\ \theta_b\sigma_b^2 - \theta\sigma_\eta^2 \end{cases} \quad (11.3)$$

$$\Sigma_o = \begin{bmatrix} 1 - \theta^2 & 0 & 0 \\ 0 & 2\sigma_b^2 & 0 \\ 0 & 0 & \frac{2\sigma_\varepsilon^2}{n-1} \end{bmatrix} \quad (11.4)$$

$$\tau_o = (\theta_b, \sigma_b^2, \sigma_\varepsilon^2)' \quad \tau_l = (\sigma_\varepsilon^2, \theta, \sigma_\eta^2)'$$

$$H_f = \begin{bmatrix} \sigma_\varepsilon^2 - \sigma_{\varepsilon*}^2 \\ H_\phi \end{bmatrix}$$

$$\frac{\partial H_f}{\partial \tau_o} = \begin{bmatrix} 0 & 0 & -1 \\ 2\theta_b\sigma_b^2 & 1 + \theta_b^2 & -n^{-1} \\ \sigma_b^2 & \theta_b & 0 \end{bmatrix}, \quad \frac{\partial H_f}{\partial \tau_l} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2\theta\sigma_\eta^2 & -(1 + \theta^2) \\ 0 & -\sigma_\eta^2 & -\theta \end{bmatrix}$$

11.0.3 Case 3: $\theta_b = 0$

We now turn to a case that was hinted at by Figures ?? and ??, namely that $\bar{\mathbf{w}}$ is AR(1). We've already established the sum of MA(1) and white noise is still MA(1). This means that $\mathbf{u}$ must be ARMA(1,1), setting up the following system of ACVF's:

$$(1 - \phi B)^{-1}b_t = (1 - \phi B)^{-1}(1 + \theta B)\eta_t + a_t$$

$$b_t = (1 + \theta B)\eta_t + (1 - \phi B)a_t$$

We calculate respective ACVF's at lags 0 and 1 to obtain:

$$H_{\theta_b} = \begin{cases} \sigma_b^2 - (1 + \theta^2)\sigma_\eta^2 - (1 + \phi^2)\sigma_a^2 \\ \theta\sigma_\eta^2 - \phi\sigma_a^2 \end{cases} \quad (11.5)$$

Note that the second equation provides a critical piece of information, namely that θ_b being zero necessitates that $\frac{\theta}{\phi} = \frac{\sigma_a^2}{\sigma_\eta^2}$. This gives rise to the following:

$$\Sigma_o = \begin{bmatrix} 1 - \phi^2 & 0 & 0 \\ 0 & 2\sigma_b^4 & 0 \\ 0 & 0 & \frac{2\sigma_\varepsilon^4}{n-1} \end{bmatrix} \quad (11.6)$$

$$\tau_o = (\phi, \sigma_b^2, \sigma_\varepsilon^2)', \quad \tau_l = (\sigma_{\varepsilon*}^2, \phi_*, \theta, \sigma_\eta^2), \quad H_f = \begin{cases} \sigma_{\varepsilon*}^2 - \sigma_\varepsilon^2 \\ \phi_* - \phi \\ H_{\theta_b} \end{cases}$$

$$\frac{\partial H_f}{\partial \tau_o} = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ -2\phi\sigma_a^2 & 1 & -(1 + \phi^2)\hat{m} \\ -\sigma_a^2 & 0 & -\phi\hat{m} \end{bmatrix}, \quad \frac{\partial H_f}{\partial \tau_l} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2\theta\sigma_\eta^2 & -(1 + \theta^2) \\ 0 & 0 & \sigma_\eta^2 & \theta \end{bmatrix}$$

In the event of an insignificant estimate for θ_b , we would refit $\mathbf{w}$ with an AR(1) process and then use H_{θ_b} to fit the remaining parameters. We synthesize the implications of these cases with the following procedure for model selection, which is performed using the parameter estimates of the converged REMARCS model.

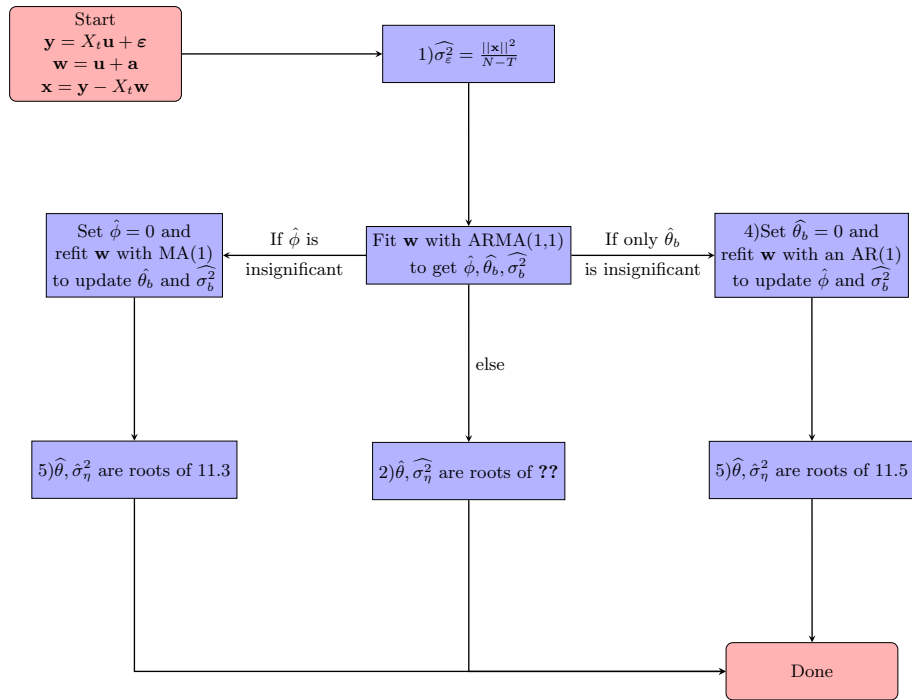


Figure 6: Decision tree for selecting parameters to include in ARMA-Noise model

12 Varying Sample Sizes

When framed in the context of repeated surveys, the requirement that the sample size at each time step is constant is unrealistic. Given that real life survey data is rarely so uniform, generalizing the method to allow for varying daily sample size is a natural goal. In examining the method as it stands, the condition of fixed sample size is only necessary in the fitting of ARMA-Noise models, as $\mathbf{w} = \mathbf{u} + \mathbf{a}$ is only an ARMA process if $\{a_t = n_t^{-1} \sum_{i=1}^{n_t}\}$ is white noise, something that at first glance is violated by varying n_t . However, we will now show how this method can be adapted to allow for varying sample sizes, so long as certain conditions are met.

13 Time-Step Averages of White Noise with Varying Sample Size

We seek to take the ARMA-Noise model and generalize it to allow for $\{n_t\}$ to vary, as is so often the case in real RCS survey data. We will now treat $\{n_t\}$ as the realizations of random variables $\{N_t\}_{t=1}^T$. What we aim to establish are the conditions under which the random variable $A_t = N_t^{-1} \sum_{i=1}^{N_t} \varepsilon_{ti}$ is still white noise.

First, let $\{N_t\}$ be i.i.d. random variables with a sample space that is a subset of the integers strictly greater than 0. The conditional random variable $A_t|N_t$ then has variance σ_ε^2/N_t . However, note that this is the variance of A_t after being conditioned on N_t , not its marginal variance.

To find $\text{Var}(A)$ we use the Law of Total Variation:

$$\text{Var}(A) = \mathbb{E}(\text{Var}(A|N)) + \text{Var}(\mathbb{E}(A|N)) \quad (13.1)$$

Now, $\text{Var}(A|N) = \sigma_\varepsilon^2 N^{-1}$, and therefore has an expectation of $\sigma_\varepsilon^2 \mathbb{E}(N^{-1})$. At the same time, the expectation of A is 0 regardless of the value of N , meaning the second term in Equation 13.1 is zero, leaving only $\text{Var}(A) = \sigma_\varepsilon^2 \mathbb{E}(N^{-1})$. It's worth identifying at this time the conditions under which $\mathbb{E}(N^{-1})$ is defined, but this will always be the case (so long as $N \geq 1$ a.s.).

Proof:

$$N \geq 1 \quad \text{a.s.}$$

$$\Rightarrow 0 < N^{-1} \leq 1 \quad \text{a.s.}$$

$$\Rightarrow 0 < \mathbb{E}(N^{-1}) \leq 1 \quad \text{a.s.}$$

Now, it is worth noting that A is only Gaussian if $\{n_t\}$ is almost surely constant. However, our claim is that $\{a_t\}$ is still white noise, just with variance $\sigma_\varepsilon^2 \mathbb{E}(N^{-1})$. To establish this, we must show that it is a zero-mean stationary process with ACVF $\gamma_A(h) = \mathbb{1}_{\{0\}}(h)\sigma^2$.

Firstly, we can quickly verify that A_t is zero mean with the Law of Total Expectation:

$$\mathbb{E}(A_t) = \mathbb{E}(\mathbb{E}(A_t|N_t)) = \mathbb{E}(0) = 0$$

Now we've already established $\text{Var}(A_t)$ to be independent of t , so we now verify that $\text{Cov}(A_r, A_s) = 0$ for $r \neq s$. The Law of Total Covariance states:

$$\text{Cov}(A_r, A_s) = \mathbb{E}(\text{Cov}(A_r, A_s|N_r, N_s)) + \text{Cov}(\mathbb{E}(A_r|N_r), \mathbb{E}(A_s|N_s))$$

Given that A_r and A_s are functions of independent random vectors $\varepsilon_r \in \mathbb{R}^{n_r}$ and $\varepsilon_s \in \mathbb{R}^{n_s}$, $\text{Cov}(A_r, A_s|N_r, N_s)$ must be zero. Having already established that both variables are zero-mean, the above equation reduces to:

$$\text{Cov}(A_r, A_s) = \mathbb{E}(0) + \text{Cov}(0, 0) = 0$$

Having shown that $\{A_t\}$ is still white noise, it now remains to estimate its variance. Though the sample sizes now vary, our estimate for σ_ε^2 hasn't changed:

$$\hat{\sigma}_\varepsilon^2 = (N - T)^{-1} \sum_{t=1}^T \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_t)^2 \quad (13.2)$$

Without requiring the distribution of N_t , σ_a^2 can be estimated with:

$$\hat{\sigma}_a^2 = \frac{\hat{\sigma}_\varepsilon^2}{T} \sum_{t=1}^T \frac{1}{N_t} \quad (13.3)$$

While we're still maintaining the independence of N_t , it's worth noting that at no point in the above reasoning was such a characteristic used in establishing A_t as white noise. Before exploring the implications of such an observation, let's look at some sample distributions for N_t that show the importance of using this estimator as opposed to more naive ones.

Example: $N_t \stackrel{d}{=} M_t + 1$, $M_t \sim \text{Poi}(\lambda)$

$N_t \geq 1$, a.s. so we know that $\mathbb{E}(N_t^{-1})$ exists. We derive it now:

$$\mathbb{E}(N^{-1}) = \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!(n+1)} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^{n+1}}{n!(n+1)} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^{n+1}}{(n+1)!} = \frac{1}{\lambda} (1 - e^{-\lambda})$$

From this we obtain the following expression for the variance of a_t :

$$\text{Var}(a_t) = \frac{\sigma_\varepsilon^2 (1 - e^{-\lambda})}{\lambda} \quad (13.4)$$

We can validate this through simulating i.i.d. samples of $N_t = M_t + 1$, with $M_t \stackrel{\text{i.i.d.}}{\sim} \text{Poi}(\lambda)$, then for each value of $t = 1, \dots, T$ we simulate N_t i.i.d. realizations of Gaussian noise ε_{ti} with variance σ_ε^2 . We set $\lambda = 2$, $\sigma_\varepsilon^2 = 10$, and $T = 1,000,000$.

Equation 13.4 yields a value of roughly $\sigma_a^2 = 4.323$ with the given parameters. Given these are simulated data, we have explicit observations of the vector $\mathbf{a}$, a luxury we won't have when fitting estimates to $X_t \mathbf{u} + \boldsymbol{\varepsilon}$.

The set $\{a_t\}$ had a sample variance of 4.324. Turning to our estimates, Equation 13.4 produced an estimate $\hat{\sigma}_a^2 \approx 4.323$, while naively estimating σ_a^2 by dividing $\hat{\sigma}_\varepsilon^2$ by the mean sample size resulted in the estimate $\frac{\hat{\sigma}_\varepsilon^2}{\bar{n}} = 3.332$. This disparity highlights the need for the estimator defined above.

Now that we have an estimate for σ_a^2 , let's see how this change in estimator affects the rest of the method.

14 ARMA-Noise Models with I.I.D. Sample Size

When sample sizes are allowed to vary over time, an ARMA-Noise model changes:

$$w_{ti} = u_t + \varepsilon_{ti}, \quad \{u_t\} \sim \text{ARMA}(\phi, \theta, \sigma_\eta^2, T), \quad \varepsilon_{ti} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\varepsilon^2)$$

$$\bar{w}_{ti} = n_t^{-1} \sum_{i=1}^{n_t} w_{ti} = u_t + a_t, \quad a_t = n_t^{-1} \sum_{i=1}^{n_t} \varepsilon_{ti}$$

$$\{n_t\} \text{ are i.i.d.} \quad \Omega(N) \subseteq \{\mathbb{Z} > 0\}$$

The individual noise variance σ_ε^2 is estimated with 13.2. We review the variance of this estimate for the purposes of estimating Σ_o (Equation 10.1)

$$\text{Var}(\hat{\sigma}_\varepsilon^2) = \text{Var} \left[(N - T)^{-1} \sum_{t \in T_2} (n_t - 1) S_t^2 \right]$$

Note that we cannot treat $(N - T)$ as a constant as we could with a fixed sample size. We leverage the Law of Total Variance to decompose the variance:

$$\text{Var}(\hat{\sigma}_\varepsilon^2) = \mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) + \text{Var}(\mathbb{E}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) \quad (14.1)$$

Focusing on the first term in Equation 14.1, we have:

$$\mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) = \mathbb{E} \left(\text{Var} \left[(N - T)^{-1} \sum_{t=1}^T \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t.})^2 | \{N_t\} \right] \right)$$

Having conditioned on $\{N_t\}$ we obtain the following:

$$\mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) = \mathbb{E} \left((N - T)^{-2} \text{Var} \left[\sum_{t=1}^T \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t.})^2 \right] \right)$$

Because of the independence within the set $\{S_t^2\}$ we get:

$$\mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \mathbb{E}\left((N-T)^{-2} \sum_{t=1}^T \text{Var}\left[\sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t.})^2\right]\right)$$

Having previously derived the variance of S_t^2 we arrive at:

$$\mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \mathbb{E}\left((N-T)^{-2} \sum_{t=1}^T 2(n_t-1)\sigma^4\right) = 2\sigma^4\mathbb{E}(N-T)^{-1}$$

Let's now turn to the second term:

$$\text{Var}(\mathbb{E}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \text{Var}(\mathbb{E}((N-T)^{-1} \sum_{t=1}^T \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t.})^2|\{N_t\}))$$

Conditioning on $\{N_t\}$, we can pass the expectation inside the first sum:

$$\text{Var}(\hat{\sigma}_\varepsilon^2|\{N_t\}) = \text{Var}\left[(N-T)^{-1} \sum_{t=1}^T \mathbb{E}\left(\sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t.})^2\right)\right]$$

With $(n_t-1)S_t^2$ having expectation $(n_t-1)\sigma_\varepsilon^2$, we arrive at:

$$\text{Var}(\mathbb{E}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \text{Var}((N-T)^{-1} \sum_t (n_t-1)\sigma_\varepsilon^2)$$

We can then pull out the σ_ε^2 :

$$\text{Var}(\mathbb{E}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \sigma_\varepsilon^4 \text{Var}((N-T)^{-1}(N-T)) = 0$$

This is consistent with the fact that, whatever $\{N_t\}$ is, $\hat{\sigma}_\varepsilon^2$ is unbiased for σ_ε^2 , therefore having no variance. We're therefore left with:

$$\text{Var}(\hat{\sigma}_\varepsilon^2) = 2\sigma^4\mathbb{E}(N-T)^{-1}$$

With only one set of sample sizes, our only estimate for $\mathbb{E}(N-T)^{-1}$ is $(N-T)^{-1}$:

$$\widehat{\text{Var}(\hat{\sigma}_\varepsilon^2)} = 2\hat{\sigma}_\varepsilon^4(N-T)^{-1}$$

In the above expression, $\hat{\sigma}_\varepsilon^4 := (\hat{\sigma}_\varepsilon^2)^2$

$$\hat{\sigma}_a^2 = \hat{m}\hat{\sigma}_\varepsilon^2, \quad \hat{m} := \left(T^{-1} \sum_{t=1}^T n_t^{-1}\right) \hat{\sigma}_\varepsilon^2 \quad (14.2)$$

With $\hat{\sigma}_\varepsilon^2$, $\hat{\sigma}_a^2$ and $\hat{\text{Var}}(\hat{\sigma}_\varepsilon^2)$ defined as above, the algorithm proposed by Wong and Miller can be once again applied, with necessary adjustments made to Σ_o and H .

15 ARMA-Noise Models With Stationary Sample Sizes

Having generalized the ARIMA-Noise model proposed by Wong to allow for varying sample size, we examine the requirement that $\{N_t\}$ be independent and identically distributed, generalizing the family of distributions to which $\{N_t\}$ can belong.

Just as it's unlikely that sample size would be exactly the same across all time steps, so too is it unlikely that daily sample sizes would be uncorrelated. It is plausible that researchers looking to maintain consistency in sample size would survey higher numbers in the event that recent previous sample sizes had been low, resulting in a negative autocorrelation. At the same time, factors affecting the ease with which data is collected might imbue the sample sizes with a positive autocorrelation.

While the introduction of temporal correlation in $\{N_t\}$ may have ramifications for $\mathbb{E}(N_t^{-1})$, the same arguments for $\{A_t\}$ being white noise still apply. An example of a natural number valued time series with autocorrelation is the Integer Valued Auto-regressive (INAR) model with binomial thinning, visualized in Figure 7 and defined by the following equation:

$$N_t = \alpha \circ B_{t-1} - \beta \circ B_{t-1} + P_t, \quad \alpha \circ B_t \stackrel{d}{=} \text{Bin}(N_{t-1}, \alpha), \quad P_t \sim \text{Poi}(\lambda) \quad (15.1)$$

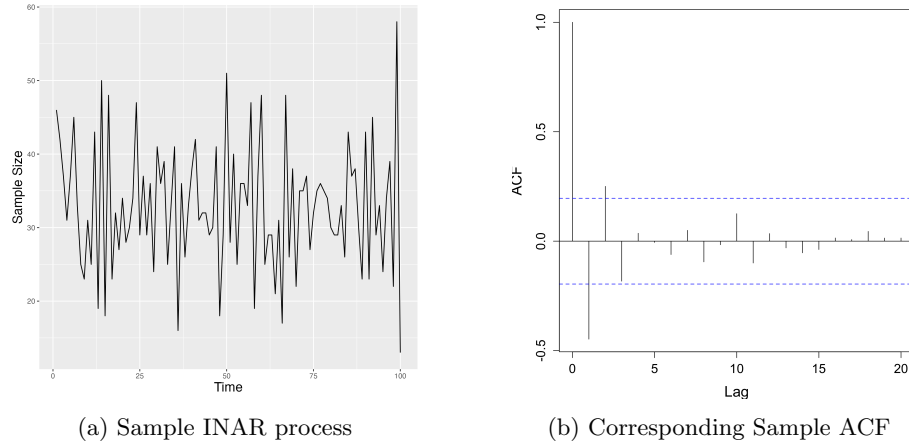


Figure 7: A realization of an INAR(2) process with parameters $\alpha = 0.3$, $\beta = 0.8$, $\lambda = 50$

Such a distribution is strictly stationary and has a constant reciprocal expectation. Were we to loosen the condition of strict stationarity to include *weakly* stationary processes, it cannot be guaranteed that $\{A_t\}$ is white noise. To illustrate this, let's define the random variables X_{odd} and X_{even} :

Thus defined, X_{odd} and X_{even} have the same first and second moments, $\mathbb{E}(X_{\text{odd}}) = \mathbb{E}(X_{\text{even}}) = 2$ and $\mathbb{E}(X_{\text{odd}}^2) = \mathbb{E}(X_{\text{even}}^2) = 5$. The time series $\{N_t\}$ defined as equal in distribution to X_{odd} for odd values of t and X_{even} for even ones is therefore weakly stationary. However, X_{odd} and X_{even} have different $(-1)^{\text{st}}$ moments:

$$\mathbb{E}(X_{\text{odd}}^{-1}) = \frac{2}{3} \quad \mathbb{E}(\text{even}^{-1}) = \frac{11}{18}$$

With N_t defined as above, the daily average A_t , while independent, has differing variance at odd and even time steps, precluding it from being white noise. It therefore becomes clear that it is not so much independence in sample sizes so much as strict stationarity that ensures $\{A_t\}$ is white noise.

We therefore can extend the procedure outlined by [4] to allow for varying sample size, so long as $\{N_t\}$ is conceivably a realization of a strictly stationary integer-valued time series. $\hat{\sigma}_a^2$ is calculated using Equation 13.3, and its asymptotic variance estimated with 13.3.

16 ARMA-Noise Processes with Non-stationary Sample Sizes

Having now established that strictly positive sample sizes being weakly stationary does not guarantee that resulting aggregates are white noise, prompting us to seek any further generalization that would allow for deterministic changes in N_t , something that would seemingly preclude resulting aggregates from being white noise. For instance, the set of sample sizes determined by $N_t = \mathbb{1}_{2\mathbb{N}}(t)n_1 + \mathbb{1}_{2\mathbb{N}-1}(t)n_2$ would immediately result in non-stationary noise. However, so far we have only established that $\mathbb{E}(N_t^{-1})$ needs to be constant across $\{t\}$. Let's then look at a distribution with constant $(-1)^{\text{st}}$ moment and a deterministic, non-constant mean with a trend:

$$\Omega(N_t) = \{1, t\}$$

We want to construct the distribution of N_t so that $\mathbb{E}(N_t^{-1}) = c \forall t$

$$\mathbb{E}(N_t^{-1}) = P(N_t = 1) + P(N_t = t)t^{-1} = c$$

Because the cardinality of $\Omega(N_t)$ is two, this expression simplifies to:

$$\mathbb{E}(N_t^{-1}) = (1 - P(N_t = t)) + P(N_t = t)t^{-1} = c$$

Defining p_t to be $P(N_t = t)$, we find:

$$p_t = \frac{t(c-1)}{1-t}$$

Since $t \geq 1$ for all t , c must be less than or equal to one. We therefore flip the direction of subtraction in the numerator and denominator:

$$p_t = \frac{t(1-c)}{t-1}$$

We can quickly double check that $\mathbb{E}(N_t^{-1})$ is in fact constant:

$$\mathbb{E}(N_t^{-1}) = 1 - p_t + \frac{p_t}{t} = 1 - \frac{t(1-c)}{t-1} + \frac{1-c}{t-1}$$

$$\mathbb{E}(N_t^{-1}) = \frac{t-1-t(1-c)+1-c}{t-1}$$

$$\mathbb{E}(N_t^{-1}) = \frac{tc-c}{t-1} = \frac{(t-1)c}{t-1} = c$$

Having now defined the distribution of N_t , let's find the expression for its mean.

$$\mathbb{E}(N_t) = 1 - p_t + p_t t = 1 + p_t(t-1) = 1 + \frac{t(1-c)}{t-1}(t-1) = 1 + t(1-c)$$

Since $c \in (0, 1)$, so too is $(1-c)$ and we get a positive linear trend for $\mathbb{E}(N_t)$. Having a mean that is linear in time, we clearly do not have a stationary process in our sequence of sample sizes. However, the aggregates of the resulting noise are uncorrelated zero-mean random variables with constant variance, and as a result are still white noise. This is visualized in Figure 8

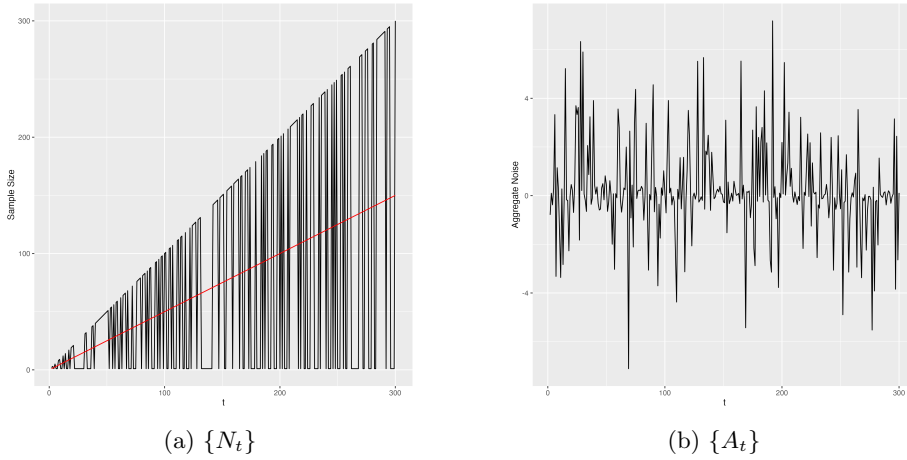


Figure 8: Sample Sizes $\{N_t = (t)M_{t+1} + 1\}$, where $M_t \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(\frac{t}{2(t-1)})$ and resulting white noise $\{A_t = N_t^{-1} \sum_{i=1}^{N_t} \varepsilon_{ti}\}$

The takeaway from all this is that a constant expected inverse of $\{N_t\}$ is the critical piece for $\{A_t\}$ to still be white noise.

We distill this observation into the following lemma:
 Let $\{\varepsilon_{ti}\}$ be i.i.d. random variables with

$$\mathbb{E}(\varepsilon_{ti}) = 0, \quad \text{Var}(\varepsilon_{ti}) = \sigma_\varepsilon^2 < \infty,$$

independent across both t and i . Let $\{N_t\}$ be a stochastic process such that

$$P(N_t \in \mathbb{N} \setminus \{0\}) = 1 \quad \text{for all } t,$$

and assume that $\mathbb{E}(N_t^{-1})$ is constant for all t .

Define

$$A_t := \frac{1}{N_t} \sum_{i=1}^{N_t} \varepsilon_{ti}.$$

Then $\{A_t\}$ is white noise with $\text{Var}(A_t) = \sigma_\varepsilon^2 \mathbb{E}(N_t^{-1})$

So long as the set of inverse sample sizes $\{n_t^{-1}\}$ is conceivably of constant expectation, the ARMA-Noise method is applicable.

17 Simulations

In order to evaluate the efficacy of the methods proposed above, individual components of the REMARCS methodology and the algorithm in its entirety were tested on simulated RCS data.

18 ARMA-Noise Model Fitting

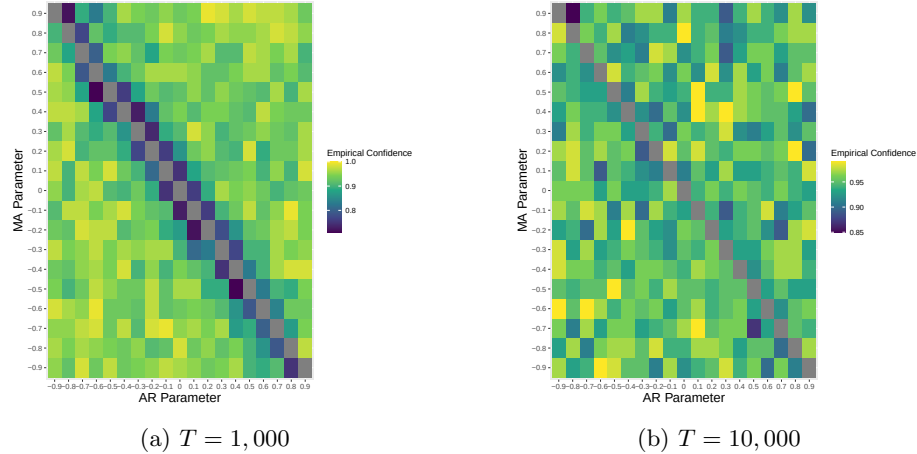


Figure 9: Rates at which point estimates for τ_o fall within confidence regions based on true values

We begin with simulations of an ARMA(1,1)-Noise Model $\mathbf{y} = X_t \mathbf{u} + \varepsilon$ parameterized by $\phi, \theta, \sigma_\eta^2, \sigma_\varepsilon^2$. Empirical confidence levels for each value-pair (ϕ, θ) are based on a hundred simulations with $\sigma_\eta^2 = \sigma_\varepsilon^2 = 1$, and $n_t = 2$. We first verify that the time series $\mathbf{w}$ defined by $w_t = \bar{\mathbf{y}}_t$, with MA parameter θ_b and innovation variance σ_b^2 , produces time series estimates with normalized covariance close to Σ_{ϕ, θ_b} . For each set of parameters, we calculate the true values of θ_b and σ_b^2 . We then fit $(\hat{\phi}, \hat{\theta}_b)$ with $\bar{\mathbf{w}}$ and construct confidence regions centered around those estimates, counting the rates at which these confidence regions capture the true parameter values. These empirical confidences are displayed in Figure 9.

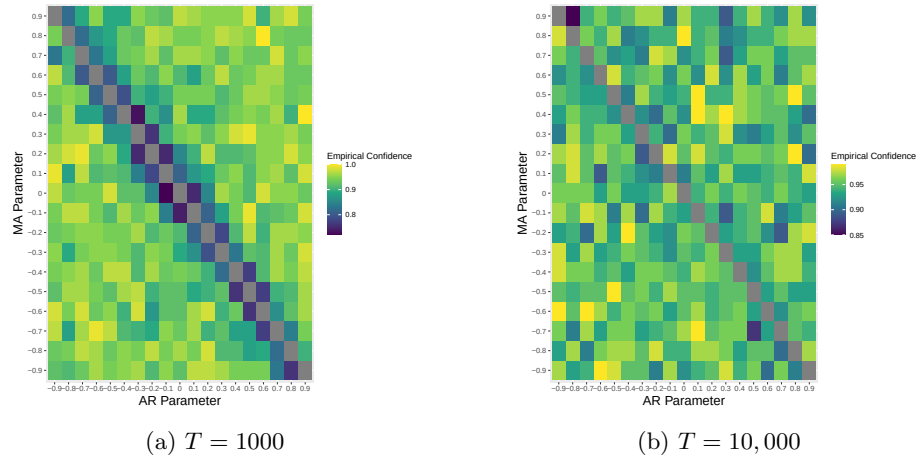


Figure 10: Rates at which point estimates for τ_l fall within confidence regions based on sample estimates

Confidence regions calculated with $\alpha = .05$ capture true values at rates close to the predicted confidence of 95%. As we are ultimately interested in estimating the parameters $\phi, \theta, \sigma_\eta^2$, we now see how well confidence regions based on applying the Delta Method to Σ_o perform. The results in Figure 10 were generated by simulating realizations of an ARMA(1,1)-Noise, estimating time series parameters ϕ and θ , then using the Delta Method to construct 95% confidence intervals.

For both observable and latent parameters we can see performance dipped for parameter combinations (ϕ, θ) close to the $\phi + \theta = 0$ line for time series of length $T = 1000$. However, we can see that performance improved for $T = 10,000$, suggesting that parameter estimates for ARMA(1,1) processes that resemble white noise take longer to converge to their asymptotic distributions.

19 REMARCS Model Fitting

In order to assess the efficacy of this method, simulations were conducted with varying combinations of ARMA parameters (ϕ, θ) . In Figure 11, we can see the proportion of 100 simulations that yielded confidence regions successfully capturing $\beta = (43, -0.001, 0.25)$ (left) and $(\phi, \theta, \sigma_\eta^2)$ (right). One observation we can make is that the estimation of time series parameters appears to be hindered close to the line defined by $\phi + \theta = 0$. This under-performance is mitigated as T grows, indicating that time series more closely resembling white noise will require larger values of T for accurate estimation.

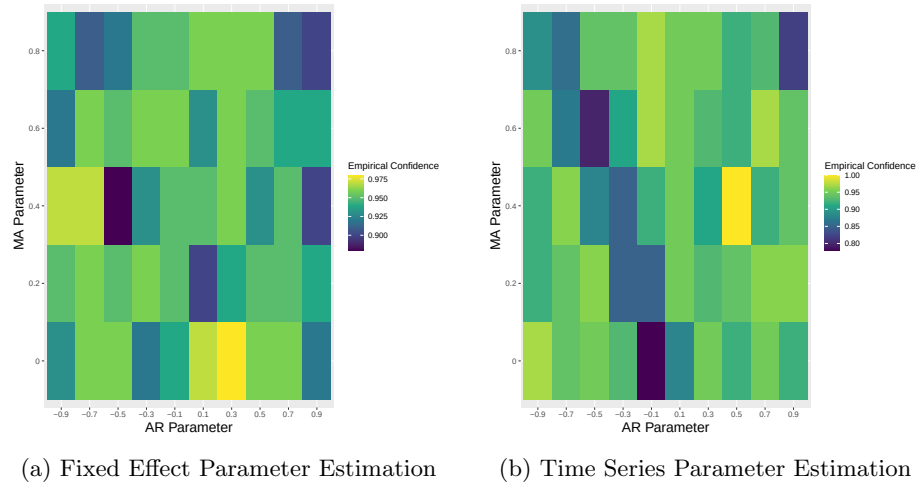


Figure 11: Empirical confidences with which model parameters fall within confidence regions generated after model convergence

19.1 Development of model parameter estimates

Now that we've seen some aggregate results displaying the efficacy of this method under different parameters, we highlight some individual realizations of specific parameter combinations and observe how the parameter estimates evolve as the algorithm progresses.

19.1.1 Case: u and w are ARMA(1,1)

We start with a single simulation of the most general case of our model, one in which all parameters are nonzero. Exact values are given below:

T	sims	ϕ	θ	σ_η^2	σ_ε^2	β_0	β_1	β_2
1000	1	0.9	-0.6	1	10	40	-0.001	2

Table 2: Parameter values for simulated REMARCS model with all nonzero parameters

We begin by displaying the individual data, along with the eventual fitted values.

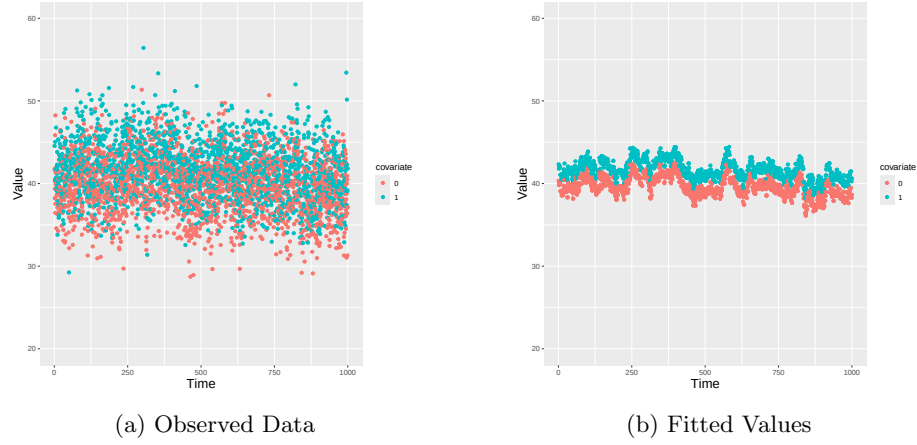


Figure 12: A realization of a REMARCS model before and after fitting

Once the convergence criterion of $|l(\hat{\xi}^{(k+1)}) - l(\hat{\xi}^{(k)})| < .001$ is satisfied, we take a look at the significance of “observable” time series parameter estimates $(\hat{\phi}, \hat{\theta}_b)$. Seeing that both $\hat{\phi}$ and $\hat{\theta}_b$ are significant in the left panel of Figure 13, we proceed to fit estimates for θ and σ_η^2 using H in Equation ???. The right panel of the same figure shows the resulting confidence region after applying the Delta Method, and we can see that the true parameters are successfully captured.

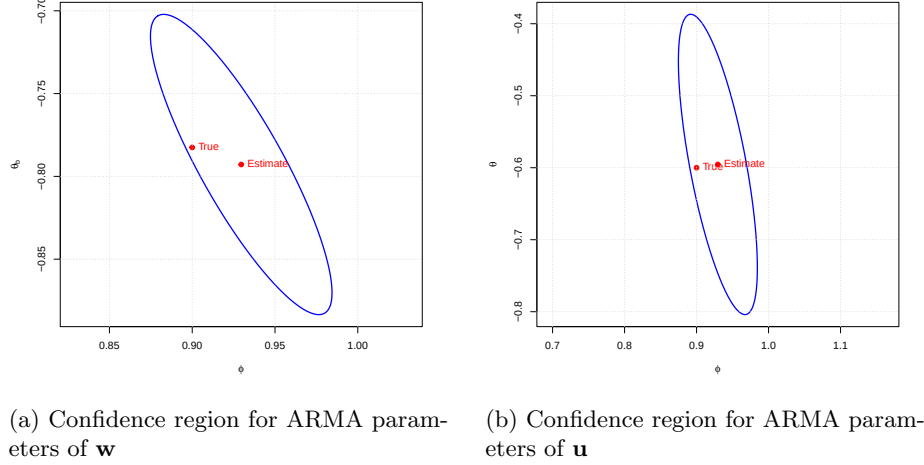


Figure 13: Confidence regions for time series parameter estimates of latent ARMA(1,1) process

In the table below we can see how properly accounting for temporal correlation helps update parameter estimates:

Parameter	β_0	β_1	β_2	σ_ε^2	ϕ	θ	σ_η^2
True Value	40.0000	-0.0010	2.0000	10.0000	0.9000	-0.6000	1.0000
Initial estimate	40.4306	-0.0016	1.9526	11.5826	0.0000	0.0000	0.0000
Final estimate	40.4352	-0.0017	1.9857	9.8552	0.9296	-0.5955	0.9482

Table 3: Parameter estimates for REMARCS model with all nonzero parameters, before and after convergence

19.1.2 Case: $\mathbf{u}$ is AR(1)

We now look at data simulated from a restricted model with the following parameters:

T	sims	ϕ	θ	σ_η^2	σ_ε^2	β_0	β_1	β_2
1000	1	0.8	0	10	10	-10	0.001	1

Table 4: Parameter values for simulated REMARCS model with a latent AR(1) process

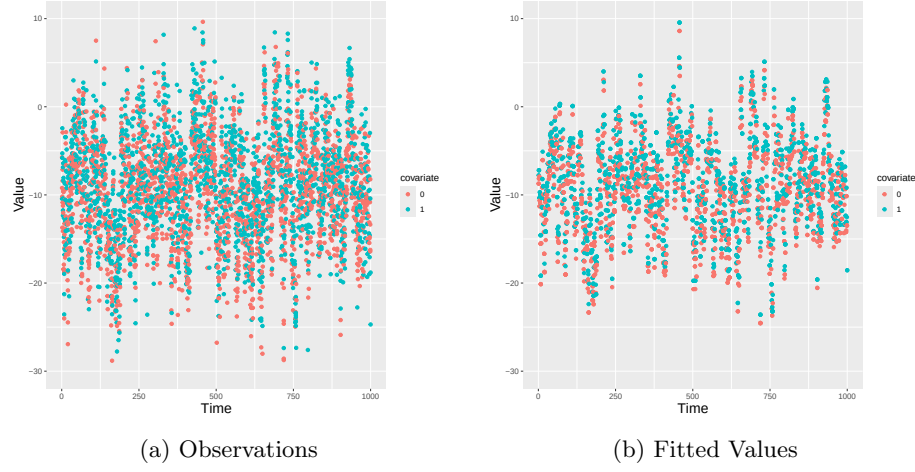


Figure 14: Observations and fitted values from a REMARCS model with a latent AR(1) time series

Parameter	β_0	β_1	β_2	σ_ε^2	ϕ	θ	σ_η^2
True Value	-10.0000	0.0010	1.0000	10.0000	0.8000	0.0000	10.0000
Initial estimate	-10.2365	0.0011	1.1015	38.6200	0.0000	0.0000	0.0000
Final estimate	-10.0172	0.0007	0.9519	9.8354	0.8130	-0.0673	10.7934

Table 5: Parameter estimates REMARCS model with latent AR(1) Time Series, before and after convergence

After having applied the general method to observations and the convergence of parameter estimates, we now perform inference on $(\hat{\phi}, \hat{\theta}_b)$. Seeing that both estimates are significant in the left side of Figure 14, we proceed as before with the most general form of H .

We then see in the right panel of Figure 14 that the resulting estimate for θ is insignificant, prompting us to repeat the estimation of σ_η^2 by using H_θ in Equation 11.1. While the standard error for $\hat{\phi}$ is unaffected, the restriction of $\hat{\theta}$ will affect $\hat{\sigma}_\eta^2$. To complete this analysis, the algorithm would be restarted, now with the restriction $\hat{\theta} = 0$.

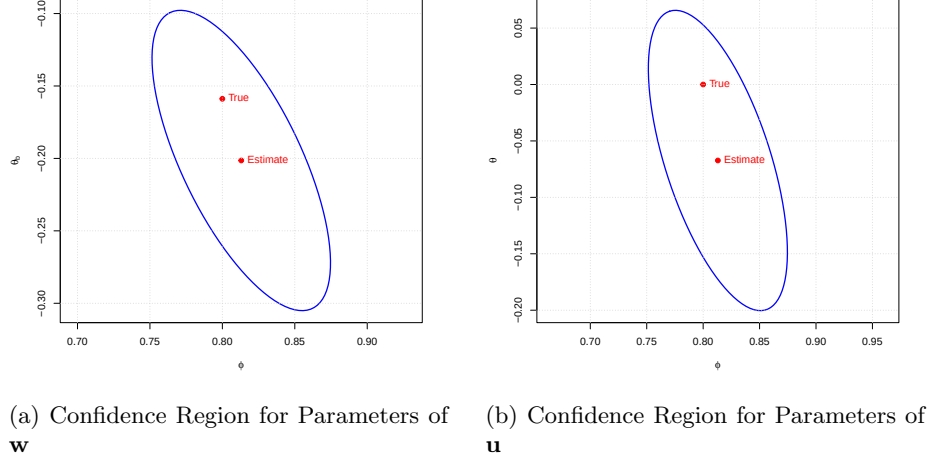


Figure 15: Confidence regions for time series parameter estimates of latent ARMA(1,1) process

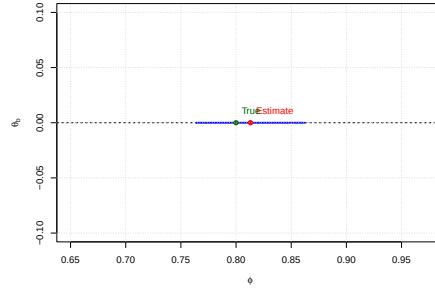


Figure 16: Confidence interval for ϕ after restricting model

19.1.3 Case: $\mathbf{u}$ is MA(1)

We now restrict $\phi = 0$ and simulate data with the following parameters:

T	sims	ϕ	θ	σ_η^2	σ_ε^2	β_0	β_1	β_2
1000	1	0	0.5	5	10	-100	0.01	3

Table 6: Parameter values for simulated REMARCS model with a latent MA(1) process

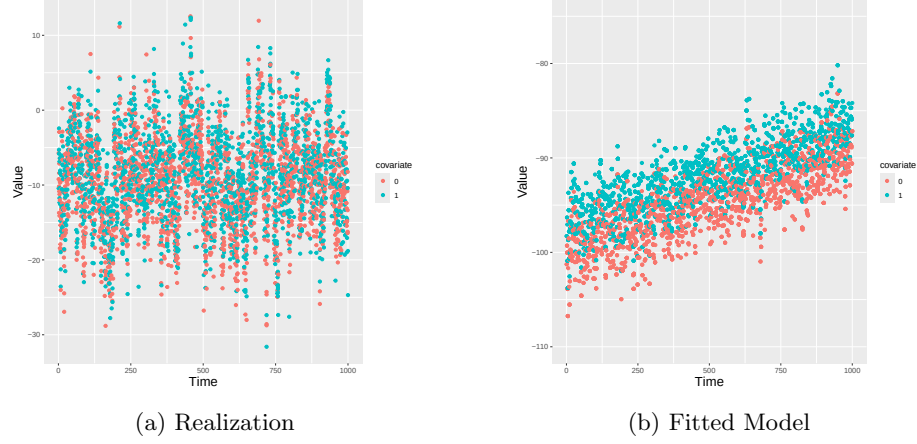


Figure 17: Simulated REMARCS Model Before and After Fitting

We then initiate the algorithm as usual and obtain the following parameter estimates:

Parameter	β_0	β_1	β_2	σ_ε^2	ϕ	θ	σ_η^2
True Value	-100.0000	0.0100	3.0000	10.0000	0.0000	0.5000	5.0000
Initial estimate	-100.0109	0.0103	2.9861	16.5249	0.0000	0.0000	0.0000
Final estimate	-100.0670	0.0104	2.9861	9.9510	-0.0816	0.5900	5.1878

Table 7: Parameter estimates for REMARCS model with latent MA(1) time series, before and after convergence

We take the final time series parameter estimates and perform inference on $\hat{\phi}$ and $\hat{\theta}_b$:

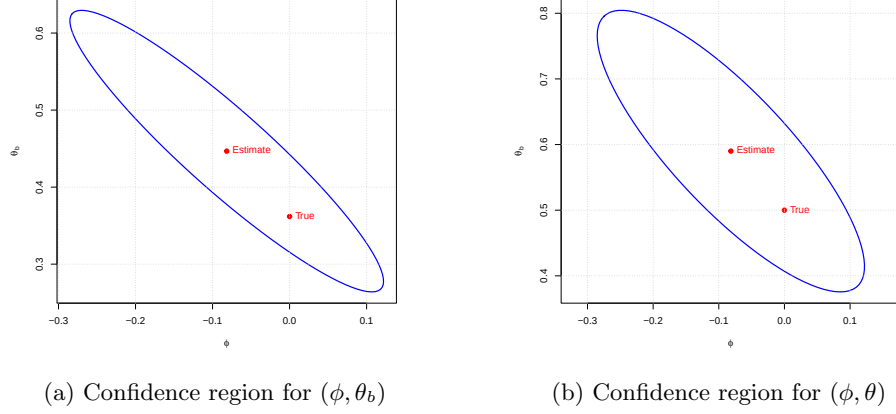


Figure 18: Confidence regions for observable and latent moving average parameters before restricting REMARCS model with $\theta_b = 0$

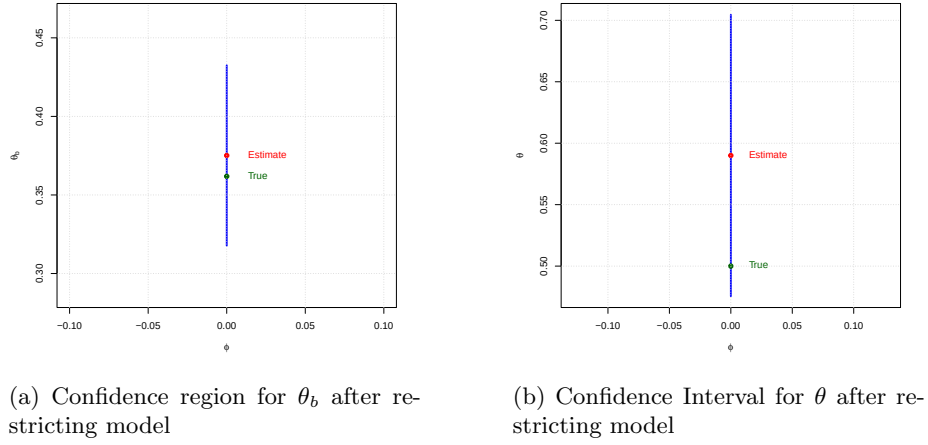


Figure 19: Confidence regions for observable and latent moving average parameters after restricting REMARCS model with $\theta_b = 0$

Note that while the point estimate $\hat{\theta}$ doesn't change much, the range of the confidence interval for θ has shrunk with the decrease in degrees of freedom.

19.1.4 Case: w is AR(1)

We now look to the special case in which aggregate noise is an AR(1) process, which we recall can only occur if $\theta/\phi = \mathbb{E}(N^{-1})\sigma_\varepsilon^2/\sigma_\eta^2$. We select parameters

for which this equality doesn't hold exactly, but is approximately true given the distribution of N_t :

T	sims	ϕ	θ	σ_η^2	σ_ε^2	β_0	β_1	β_2
1000	1	0.5	0.5	5.25	50	20	-0.001	2

Table 8: Parameter values for simulated REMARCS model with latent AR(1) process

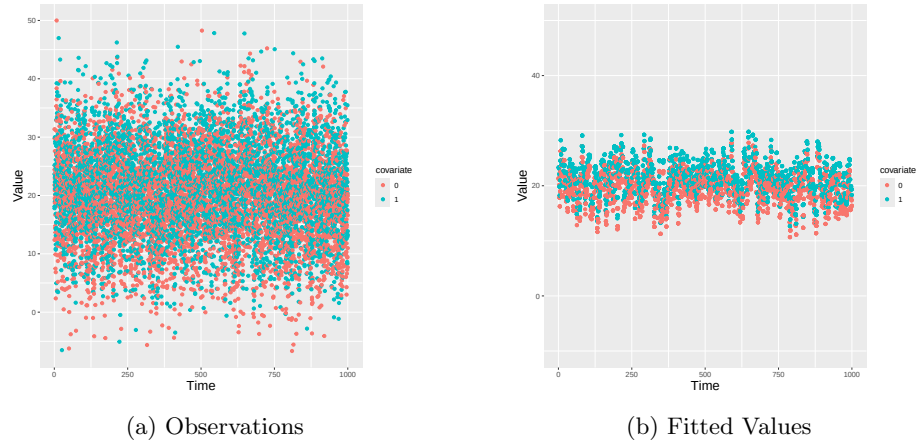


Figure 20: Realization of a REMARCS Model in which the sum of the latent time series and aggregated cross-sectional noise is an AR(1) process

After fitting a REMARCS model to the data we obtain parameter estimates shown in Table 9 and fitted values in Figure 20.

Parameter	β_0	β_1	β_2	σ_ε^2	ϕ	θ	σ_η^2
True Value	20.0000	-0.0010	2.0000	50.0000	0.5000	0.5000	5.2500
Initial estimate	19.6845	-0.0007	2.0362	61.4985	0.0000	0.0000	0.0000
Final estimate	19.7719	-0.0008	1.9851	49.7772	0.4996	0.4962	5.1336

Table 9: Parameter estimates for REMARCS model with $\theta_b \approx 0$, before and after convergence

Note that while the point estimate for θ given by Table 9 is accurate, the confidence region thereof (Figure 21) practically spans all of $(0, 1)$, with which all that can be said is that θ is positive.

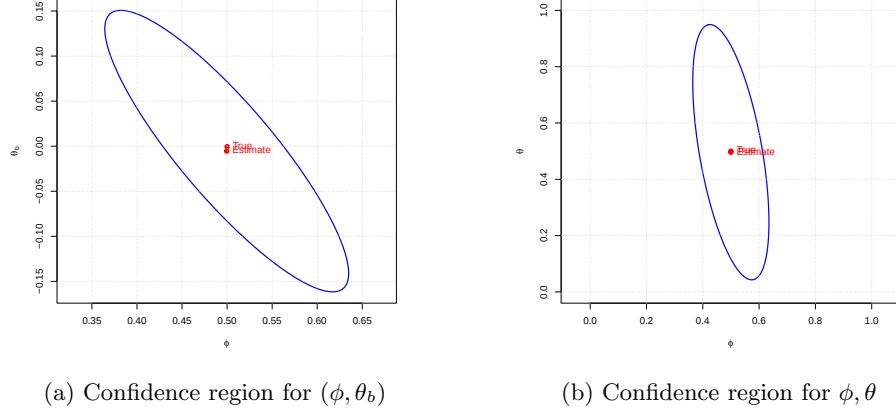


Figure 21: Inference on ARMA parameters of REMARCS model in which $\theta_b = 0$ with restricted model

Note in the left panel of the same figure we see that the estimate for θ_b is insignificant, prompting us to refit $\mathbf{w}$ with an AR(1) model and use H_{θ_b} (Equation 11.5) to construct a confidence region for $\hat{\theta}$, the products of which are shown in Figure 22:

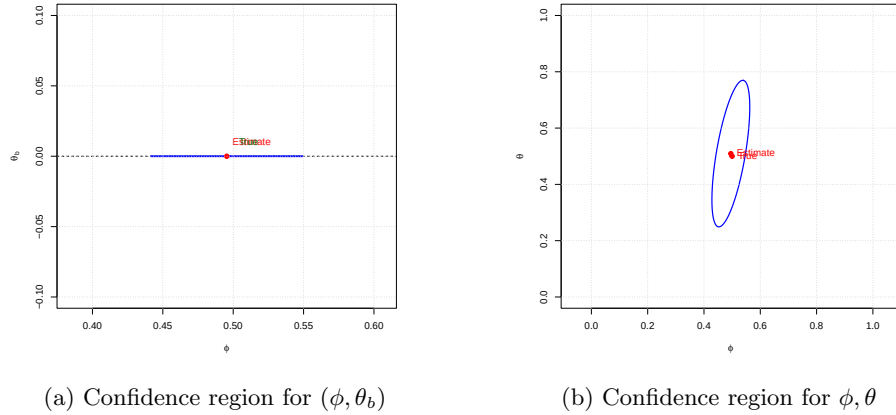


Figure 22: Inference on ARMA parameters of REMARCS model in which $\theta_b = 0$ with restricted model

We now obtain a confidence interval for θ markedly smaller than before. Another change of note is that the correlation between $\hat{\phi}$ and $\hat{\theta}$ has gone from negative to positive. This makes sense given that the ratio between the two now has an added restriction.

Now that our algorithm has demonstrated promising results on simulated data, we observe how it performs on real RCS survey data.

20 Discussion

This dissertation develops a framework for analyzing high-frequency repeated cross-sectional survey data that exhibit temporal correlation while containing individual-level covariates that are time independent. It does so by introducing a latent ARMA process, inducing an overall ARMA-Noise residual structure, and recursively fitting parameter estimates with GLS and ARMA-Noise methods. In order to accommodate survey data with varying daily sample size, this research furthered ARMA-Noise modeling by relaxing the preexisting condition that sample sizes be constant. Results provide insight into how temporal correlation interacts with cross-sectional noise, particularly how not properly accounting for this interaction can obfuscate results and hinder proper inference. Given how large RCS datasets can be, computational methods were tailored to the specific characteristics of this model by leveraging the relationship between the spectral decompositions of the various covariance matrices involved. The exploration of these models has produced insight into the limitations of their analysis and yielded several interesting directions for further research.

A result central to this work is that parameter estimates for a latent ARMA process will be fundamentally flawed if fit to aggregate residuals, as the presence of additional white noise introduces bias in estimates for not only the variance parameter but the moving average parameter as well. Such an impact is tractable, however, and can be reverse engineered to obtain asymptotically unbiased estimates for the true time series parameters. The relation between the observable and latent time series must be incorporated into the analysis of time series estimate. This impact can change the order of the ARMA process on top of the parameters themselves. For instance, in the case that the latent time series is actually an AR(1) process, the observed aggregate errors will be ARMA(1,1) with a moving average parameter of opposite sign as that of the autoregressive parameter. Modeling the noisy time series as an AR(1) process would prevent a consistent estimate for ϕ from being obtained. Simulation results reinforce this conclusion. We see that as the relative variance of the additional noise increases, estimates for ϕ obtained by fitting an AR(1) model become increasingly biased, while those fit with an ARMA(1,1) model remain accurate.

This work provides a viable method for analyzing datasets such as the mVAM surveys, elucidating individual-level effects by properly accounting for temporal shocks that would otherwise obfuscate them. Such a method avoids the information loss inherent to the aggregate methods that are currently commonplace.

Furthermore, failure to account for temporal correlation when modeling repeated cross-sectional data can lead to invalid inference. At the same time, flattening the data to be analyzed into aggregates results in significant information loss at the individual level. In particular, in the event that individual-level

variables like education level or income source see little variance in their prevalence at the time-step level, fitting aggregated data using a time series with covariates could easily obscure an otherwise significant effect.

This method is not without limitations, particularly given its reliance on asymptotic distributions. As such confidence regions underperform for low values of T , particularly in boundary cases like $\phi, \theta \approx \pm 1$ and $\phi + \theta \approx 0$, as the resulting covariance structure is close to that of white noise. The imposition of an ARMA order no greater than $(1, 1)$ is certainly limiting, as it precludes REMARCS Models with all but three types of latent ARMA processes from being properly fit.

There are a number of natural extensions that arise from this work, particularly generalizing the individual level of the model to allow for random effects. Relaxing the conditions that $p, q \leq 1$ and generalizing the latent time series to allow for any general ARMA process is also a natural question. Given that survey data so often tracks binary output variables, relaxing the Gaussian condition is of further interest. Since confidence intervals are currently based on asymptotic distributions, the derivation of small-sample corrections is warranted.

References

- [1] George E. P. Box and Gwilym M. Jenkins. *Time Series Analysis: Forecasting and Control*. Holden-Day, San Francisco, CA, second edition, 1976.
- [2] Peter J. Brockwell and Richard A. Davis. *Time Series: Theory and Methods*. Springer Series in Statistics. Springer, New York, 2 edition, 1991.
- [3] Thomas A. DiPrete and David B. Grusky. Multilevel analysis of trends with an application to the study of income inequality in the united states. *American Sociological Review*, 55(3):438–451, 1990.
- [4] Wing keung Wong and Robert B. Miller. Repeated time series analysis of arima-noise models. *Journal of Business & Economic Statistics*, 8(2):243–250, 1990.
- [5] Lucia Modugno, Silvia Cagnone, and Simone Giannerini. A multilevel model with autoregressive components for the analysis of tribal art prices. *Journal of Applied Statistics*, 42(10):2141–2158, 2015.
- [6] A. J. Scott and T. M. F. Smith. Analysis of repeated surveys using time series methods. *Journal of the American Statistical Association*, 69(347):674–678, 1974.

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